Class 17 Ruling in or out limit cycles

Activity

Teams 1 & 2, post photos of your work to the class Google Drive (see Canvas for link). Make a folder for today's class if one doesn't exist yet.

1. (Constructing a trapping region using polar)

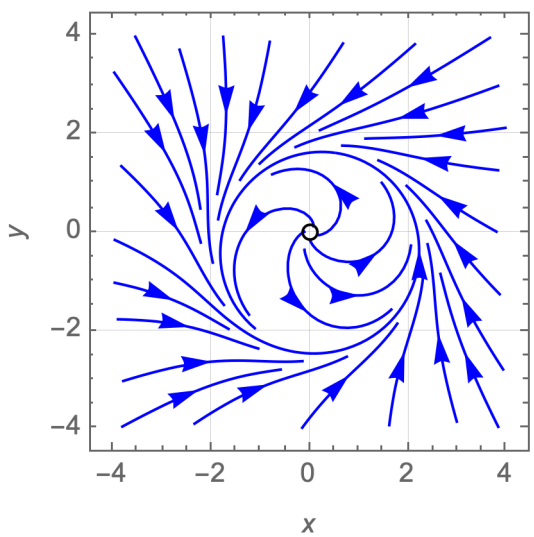
   Consider the system

$$\dot{r} = r(2 - \sin\theta - r)$$
$$\dot{\theta} = 1.$$

   (a) Show that the curve $r = 2 - \sin\theta$ is not a phase curve of the system.
   Trajectories that start on this curve will leave it.

   (b) For $r = 4$, identify the minimum and maximum possible values of $\dot{r}$ that could occur.

   (c) Construct a trapping region that satisfies the conditions of the Poincaré-Bendixson theorem.



2. In the problem above, if it had not been possible to construct a trapping region, what would you have learned about the system from the Poincaré-Bendixson theorem?

3. (Find a trapping region by converting to polar?)

   Consider the system

$$\dot{x} = y$$
$$\dot{y} = -x - y(x^2 + y^2 - 1).$$

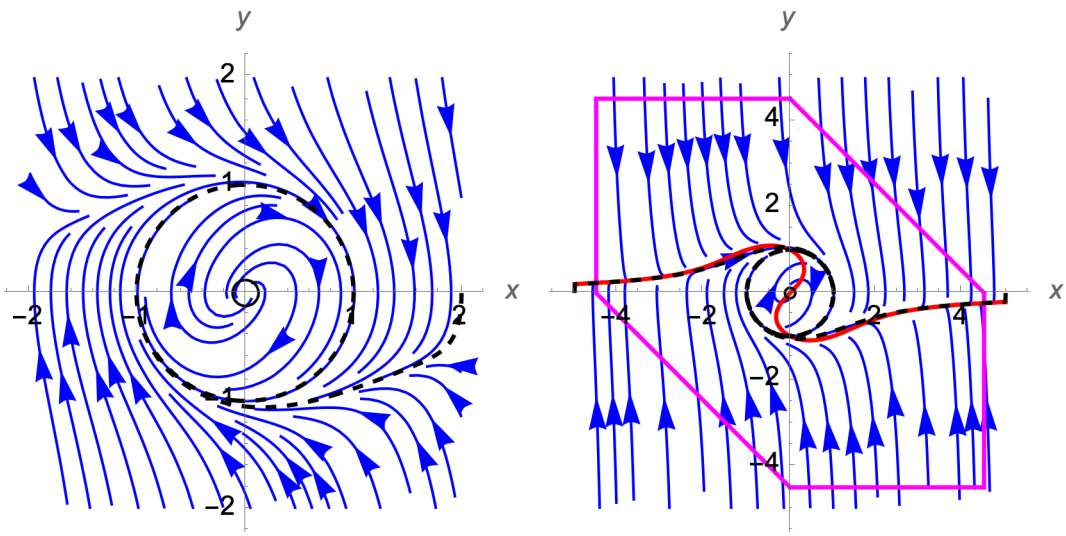
   (a) Transform the system to polar coordinates.

   (b) Show that there is an invariant set at $r = \sqrt{x^2 + y^2} = 1$.

   (c) In polar coordinates the natural trapping regions are circles. If you try to create a circular outer trapping region, what issue arises?

(d) Consider the magenta box on the right, below. Identify the mathematical work you need to do to show that it is a trapping region.

If time permits, show it is a trapping region.



4. (Ruling out closed orbits) Let

$$\dot{x} = y$$
$$\dot{y} = x - x^3 - \mu y, \quad \mu \geq 0.$$

This system is known as the unforced Duffing oscillator. (Duffing published work on this kind of equation in 1918, and it became a popular model in the 1970s). The μy term is a damping term.

(a) This system can also be written as $\ddot{x} = x - x^3 - \mu y$. How does this differ from the similar form that we have seen for a conservative system?

(b) Use Bendixson's criterion to show that the system has no closed orbits for $\mu > 0$.

5. (7.2.10) Find a and b such that $V(x, y) = ax^2 + by^2$ is a Liapunov function for

$$\dot{x} = y - x^3$$
$$\dot{y} = -x - y^3$$